

GRASSWORKS: Analysis of Bauer et al. (submitted) Functional traits of grasslands: Community weighted mean of specific leaf area (SLA) per plot (esy4)

MB

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NOTE: To compare different models, you only have to change the models in the section ‘Load models’

Preparation

Protocol of data exploration (Steps 1-8) used from Zuur et al. (2010) Methods Ecol Evol DOI: 10.1111/2041-210X.12577

```
library(here)
library(tidyverse)
library(ggbeeswarm)
library(patchwork)
```

```
library(DHARMA)
library(emmeans)
```

Packages

```
sites <- read_csv(
  here("data", "processed", "data_processed_sites_esy4.csv"),
  col_names = TRUE, na = c("na", "NA", ""), col_types = cols(
    .default = "?",
    eco.id = "f",
    region = col_factor(levels = c("north", "centre", "south"), ordered = TRUE),
    site.type = col_factor(
      levels = c("positive", "restored", "negative"), ordered = TRUE
    ),
    fertilized = "f",
    obs.year = "f",
    hydrology = "f",
    mngm.type = "f"
  )
) %>%
mutate(
  esy4 = fct_relevel(esy4, "R", "R22", "R1A"),
  eco.id = factor(eco.id)
) %>%
rename(y = cwm.abu.sla)
```

Load data

Statistics

Data exploration

Means and deviations

```
Rmisc::CI(sites$y, ci = .95)
```

```
##      upper      mean      lower
## 241.2086 238.5490 235.8893
```

```
median(sites$y)
```

```
## [1] 238.45
```

```
sd(sites$y)
```

```
## [1] 33.75003
```

```
quantile(sites$y, probs = c(0.05, 0.95), na.rm = TRUE)
```

```
##      5%      95%
## 178.73 293.08
```

```
sites %>% count(eco.id)
```

```
## # A tibble: 3 x 2
##   eco.id      n
```

```
##   <fct>  <int>
## 1 654      203
## 2 664      203
## 3 686      215
```

```
sites %>% count(site.type)
```

```
## # A tibble: 3 x 2
##   site.type      n
##   <ord>      <int>
## 1 positive    102
## 2 restored    401
## 3 negative    118
```

```
sites %>% count(esy4)
```

```
## # A tibble: 3 x 2
##   esy4      n
##   <fct> <int>
## 1 R      330
## 2 R22    210
## 3 R1A     81
```

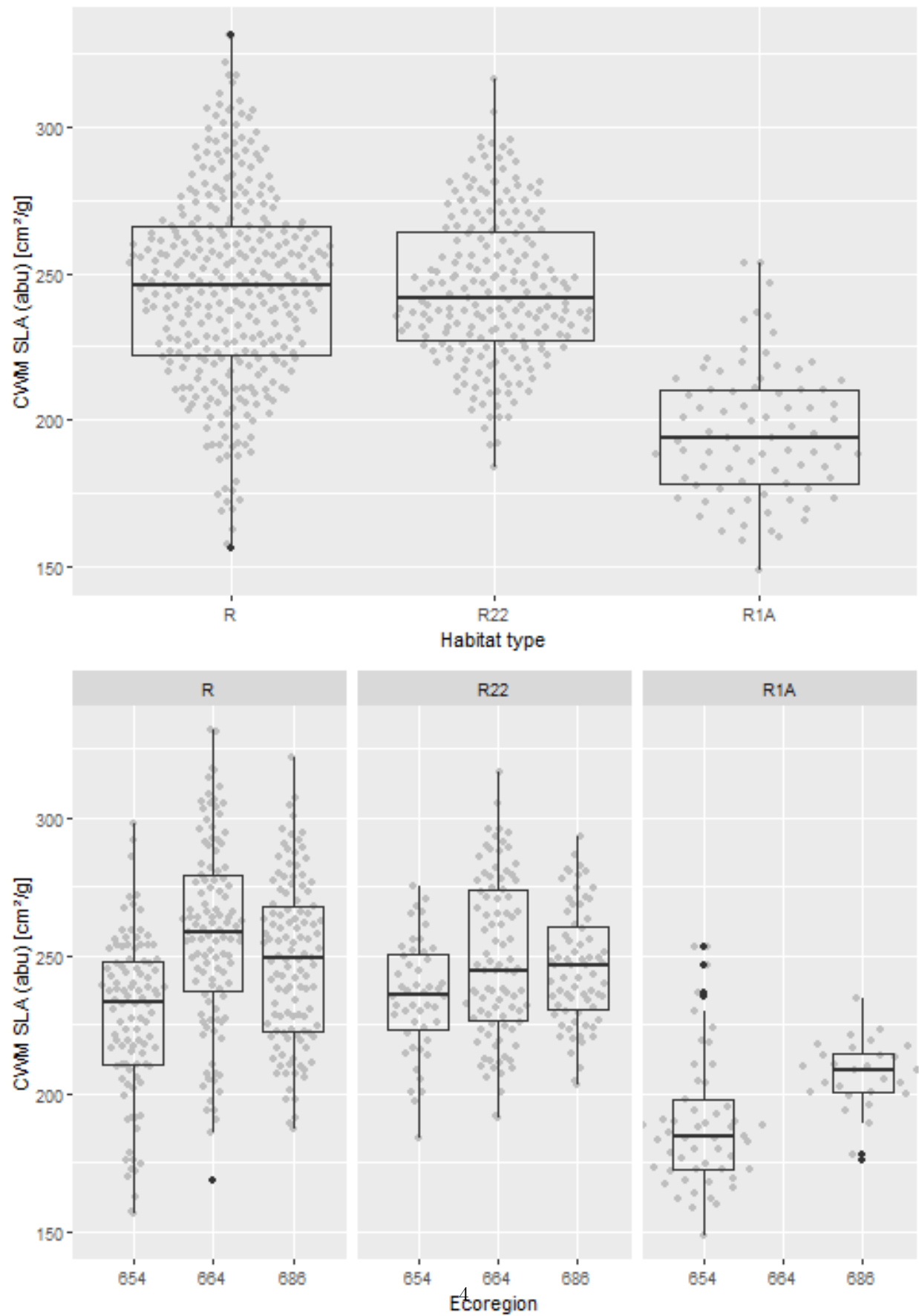
```
sites %>% count(esy4, eco.id)
```

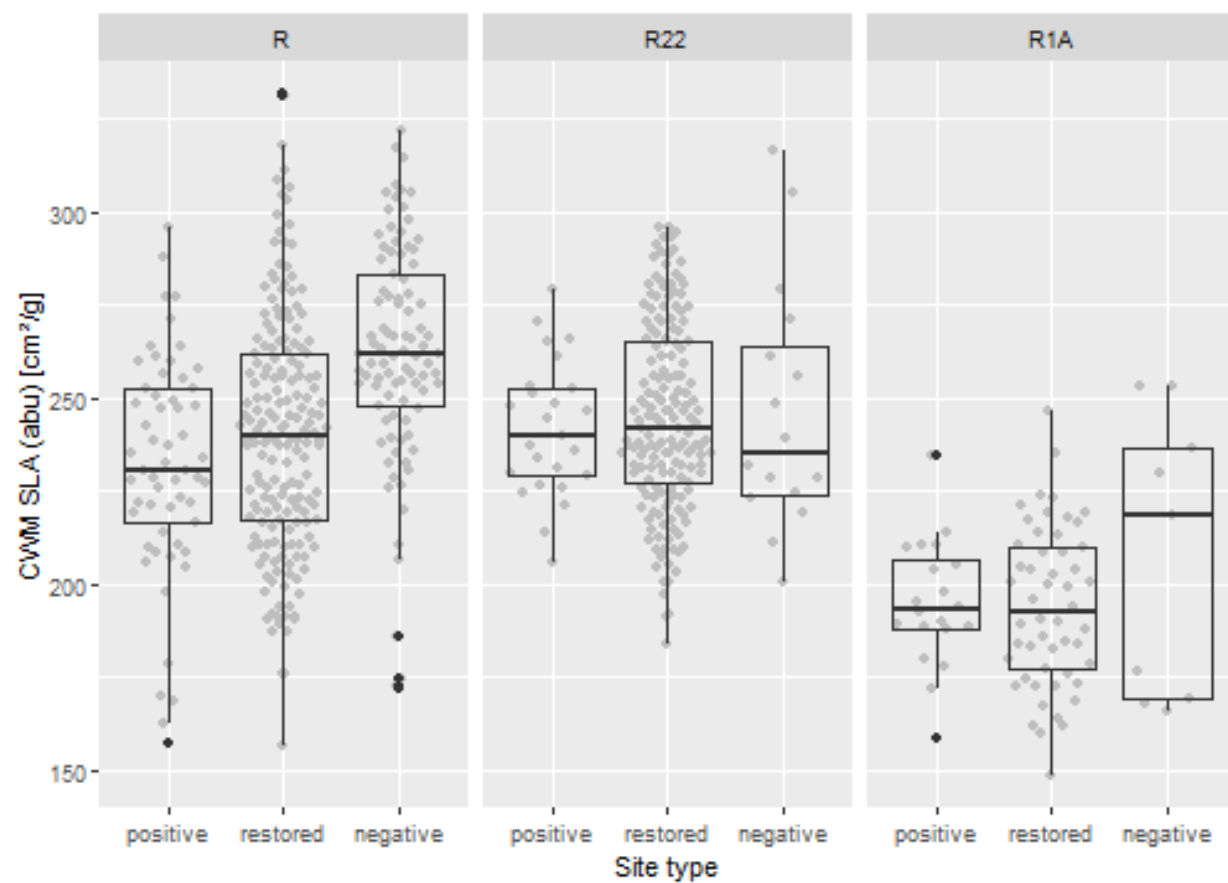
```
## # A tibble: 8 x 3
##   esy4 eco.id      n
##   <fct> <fct> <int>
## 1 R      654     102
## 2 R      664     112
## 3 R      686     116
## 4 R22    654      48
## 5 R22    664      91
## 6 R22    686      71
## 7 R1A    654      53
## 8 R1A    686      28
```

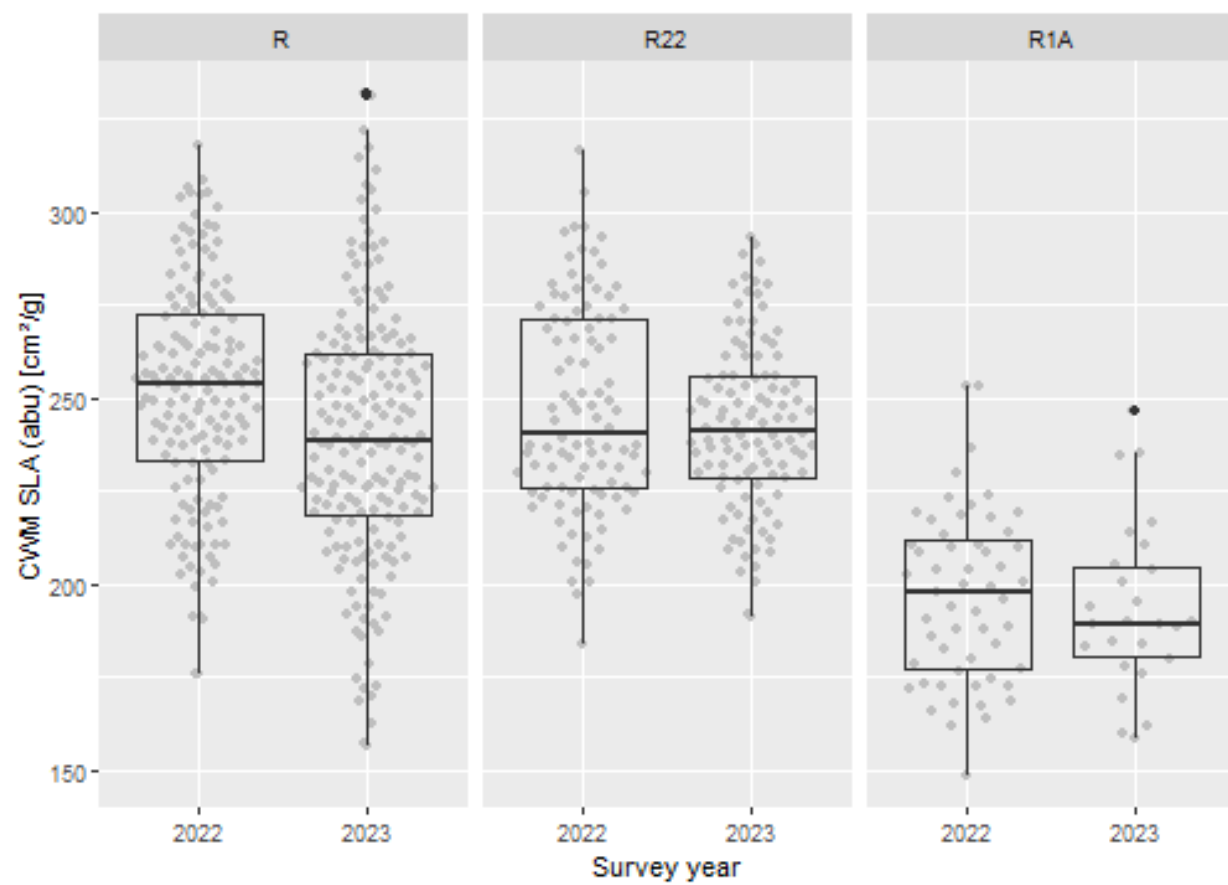
```
sites %>% count(esy4, site.type)
```

```
## # A tibble: 9 x 3
##   esy4 site.type      n
##   <fct> <ord>      <int>
## 1 R      positive     57
## 2 R      restored    180
## 3 R      negative     93
## 4 R22    positive     25
## 5 R22    restored    169
## 6 R22    negative     16
## 7 R1A    positive     20
## 8 R1A    restored     52
## 9 R1A    negative      9
```

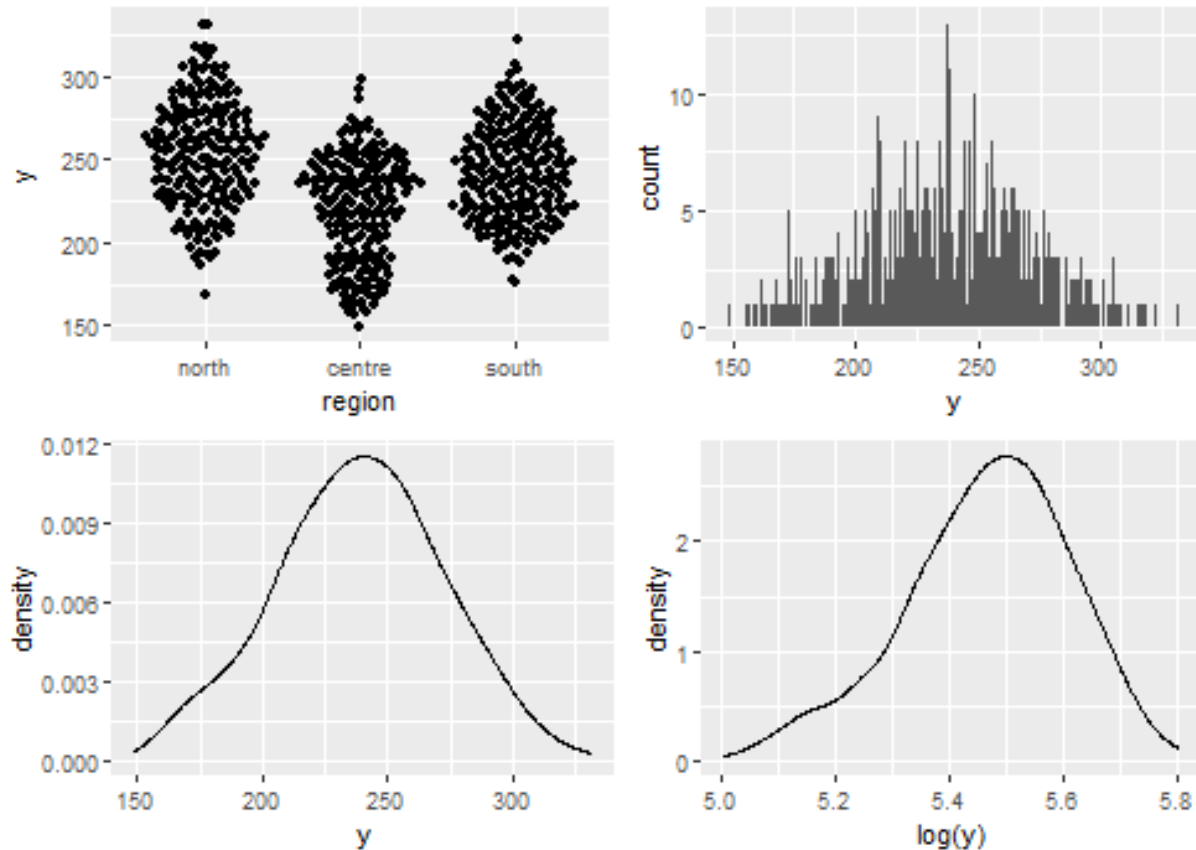
Graphs of raw data (Step 2, 6, 7)







Outliers, zero-inflation, transformations? (Step 1, 3, 4)



Check collinearity part 1 (Step 5)

Exclude $r > 0.7$ Dormann et al. 2013 Ecography DOI: 10.1111/j.1600-0587.2012.07348.x

```
# sites %>%
#   select(where(is.numeric), -y, -starts_with("cum. ")) %>%
#   GGally::ggpairs(
#     lower = list(continuous = "smooth_loess")
#   ) +
#   theme(strip.text = element_text(size = 7))

# -> no continuous variables
```

Models

NOTE: Only here you have to modify the script to compare other models

```
load(file = here("outputs", "models", "model_sla_esy4_1.Rdata"))
load(file = here("outputs", "models", "model_sla_esy4_3.Rdata"))
m_1 <- m1
m_2 <- m3

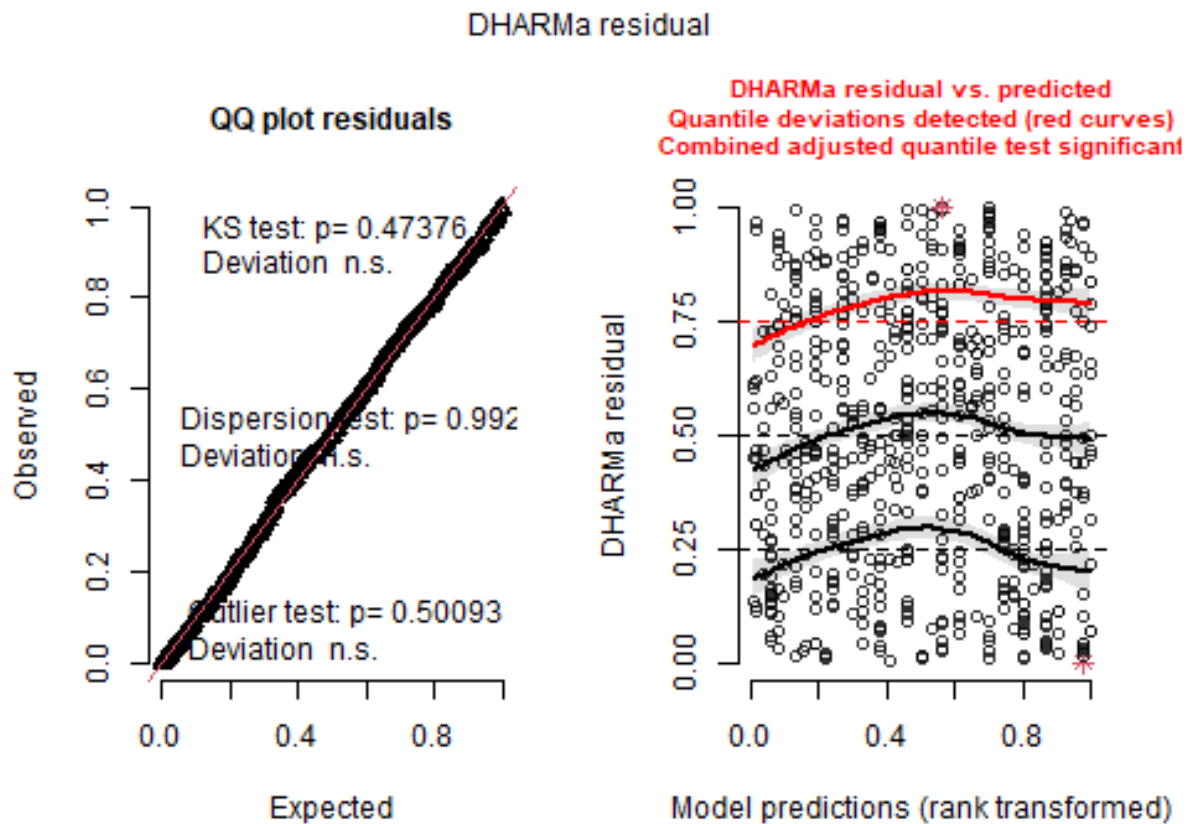
m_1@call
## lmer(formula = y ~ esy4 * (site.type + eco.id) + obs.year + (1 |
##   id.site), data = sites, REML = FALSE)
```

```
m_2@call
## lmer(formula = y ~ esy4 * site.type + eco.id + obs.year + hydrology +
##       (1 | id.site), data = sites, REML = FALSE)
```

Model check

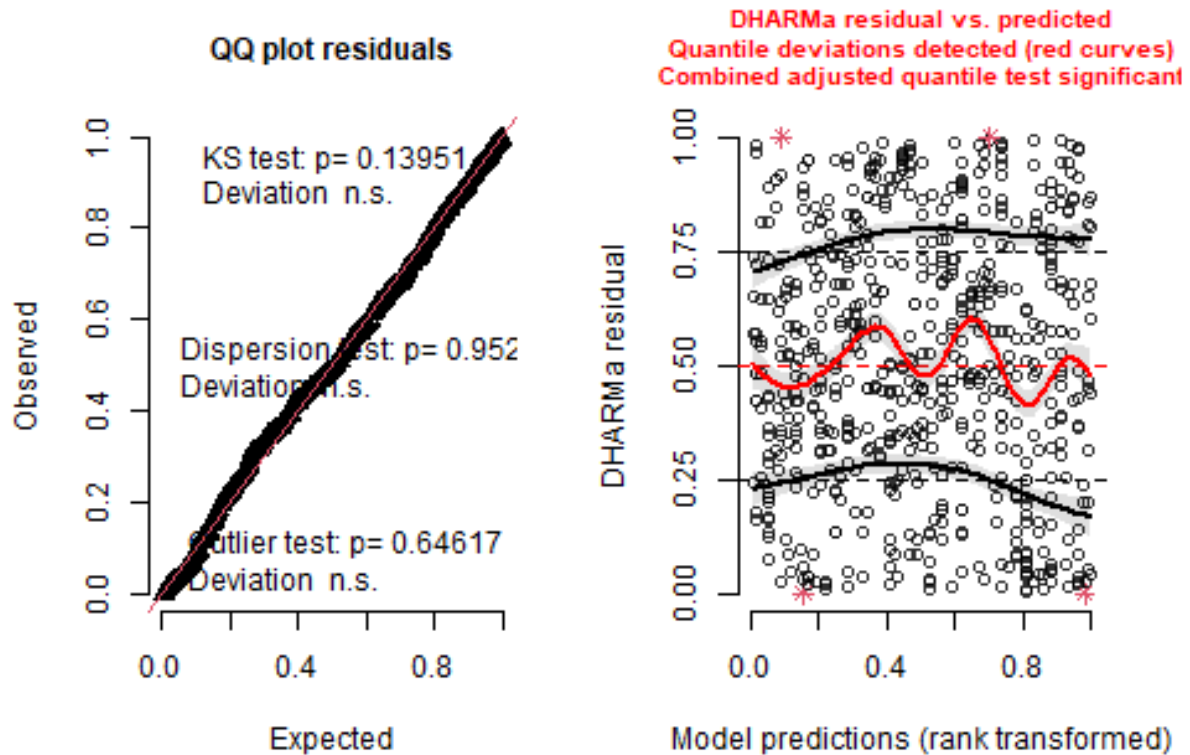
DHARMA

```
simulation_output_1 <- simulateResiduals(m_1, plot = TRUE)
```

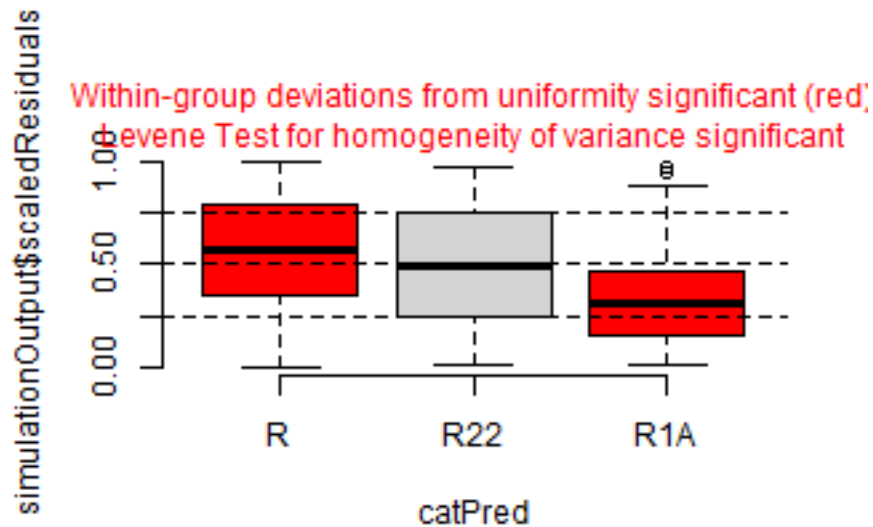


```
simulation_output_2 <- simulateResiduals(m_2, plot = TRUE)
```

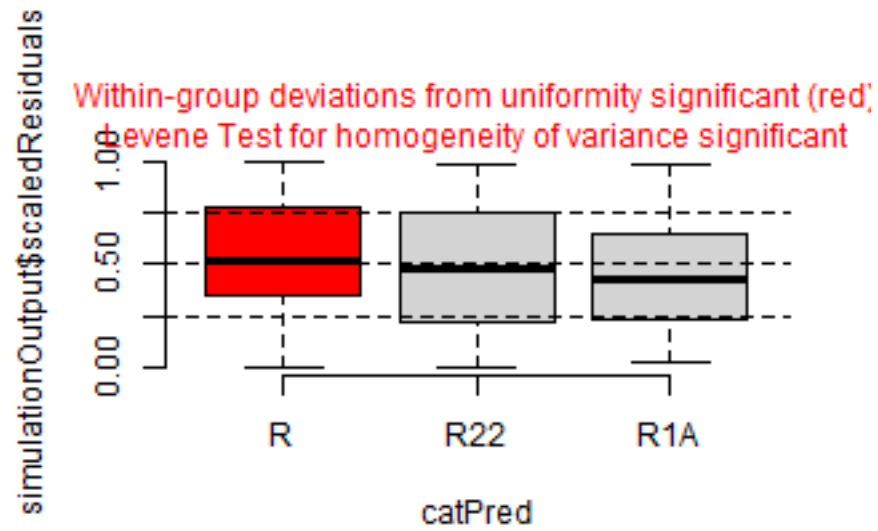

DHARMa residual



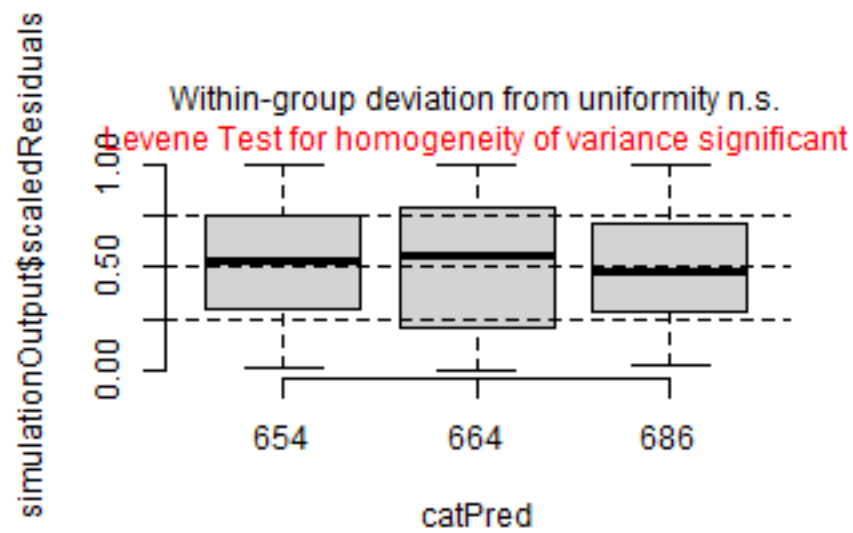
```
plotResiduals(simulation_output_1$scaledResiduals, sites$esy4)
```



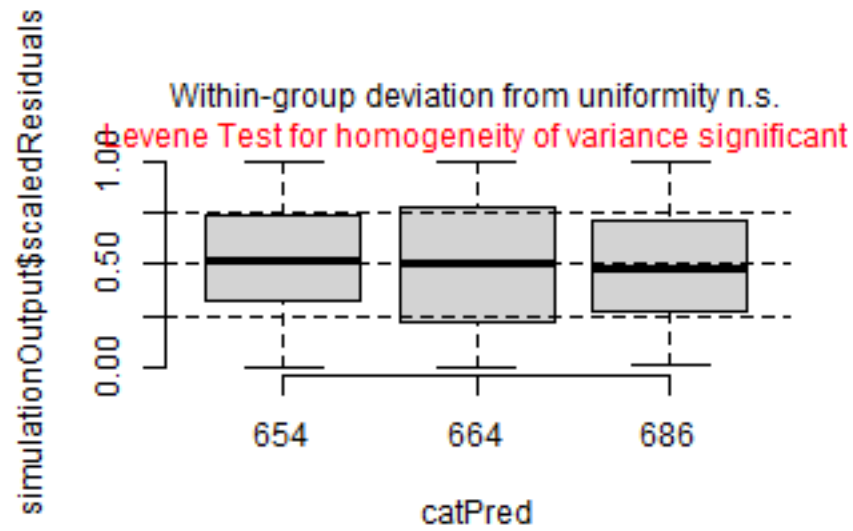
```
plotResiduals(simulation_output_2$scaledResiduals, sites$esy4)
```



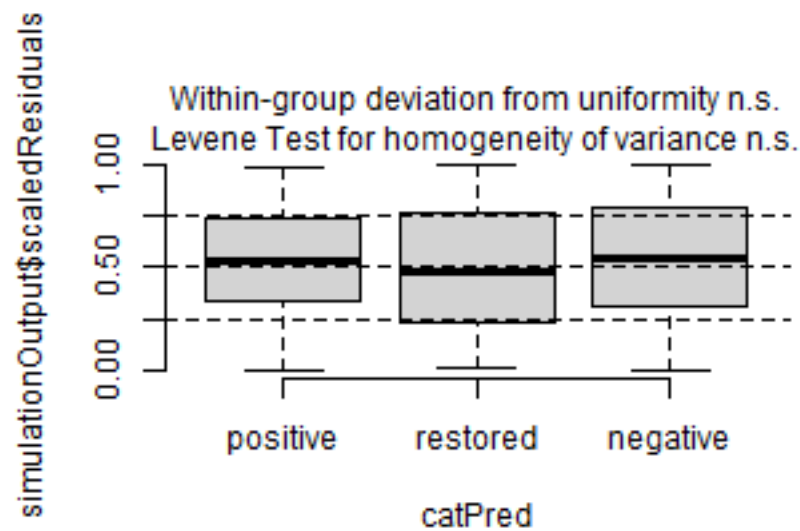
```
plotResiduals(simulation_output_1$scaledResiduals, sites$eco.id)
```



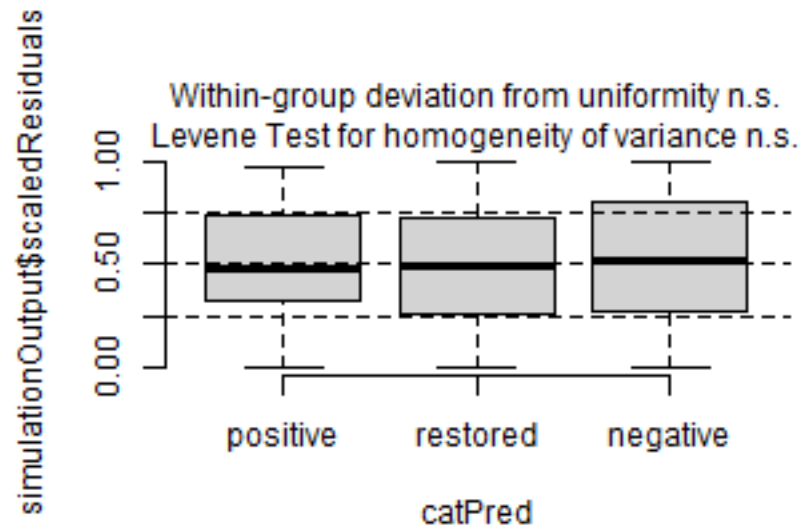
```
plotResiduals(simulation_output_2$scaledResiduals, sites$eco.id)
```



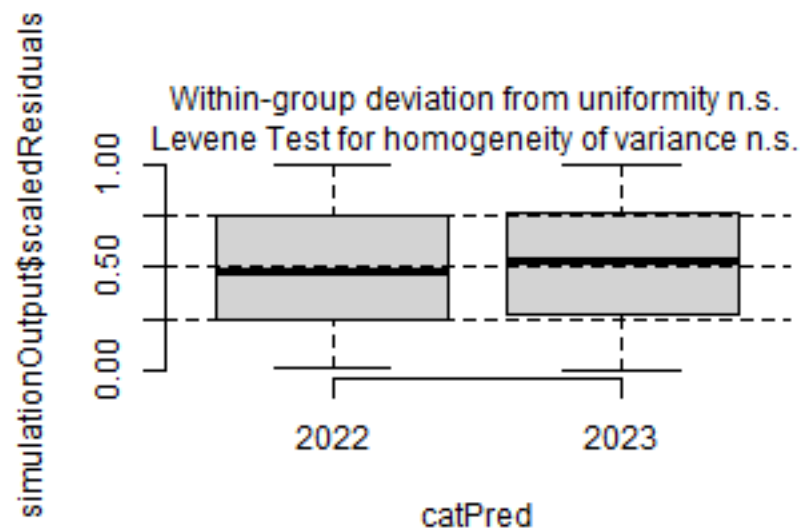
```
plotResiduals(simulation_output_1$scaledResiduals, sites$site.type)
```



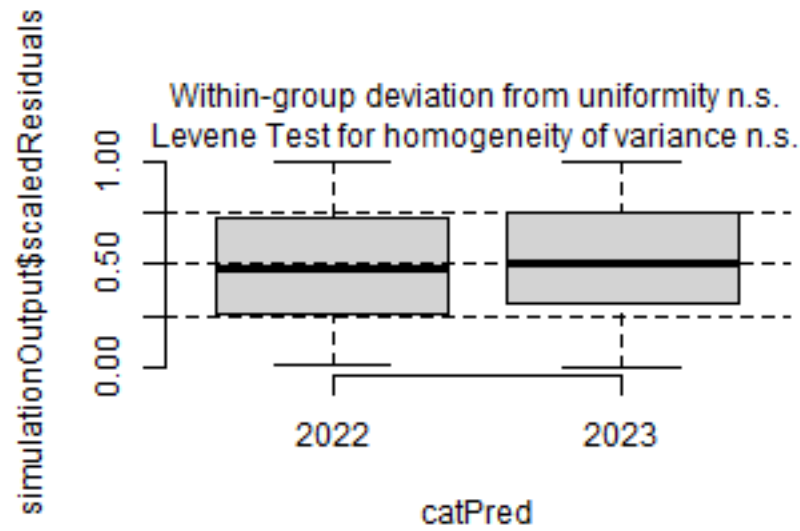
```
plotResiduals(simulation_output_2$scaledResiduals, sites$site.type)
```



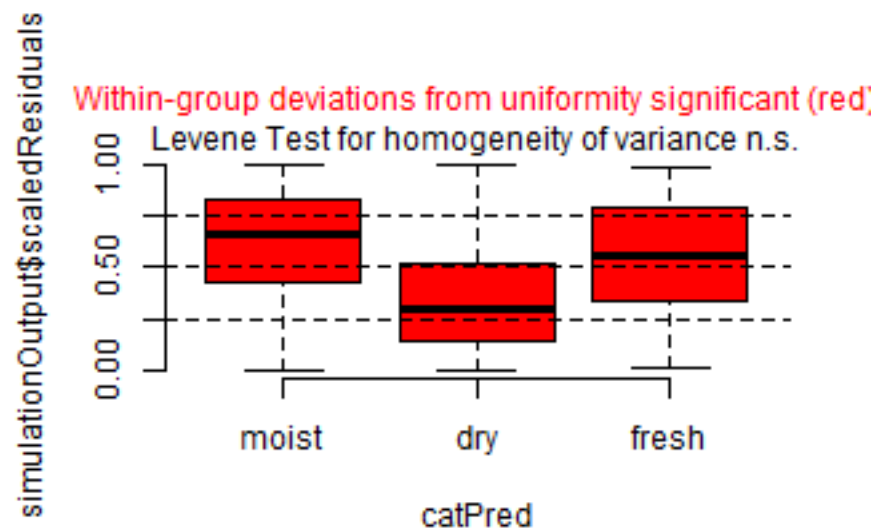
```
plotResiduals(simulation_output_1$scaledResiduals, sites$obs.year)
```



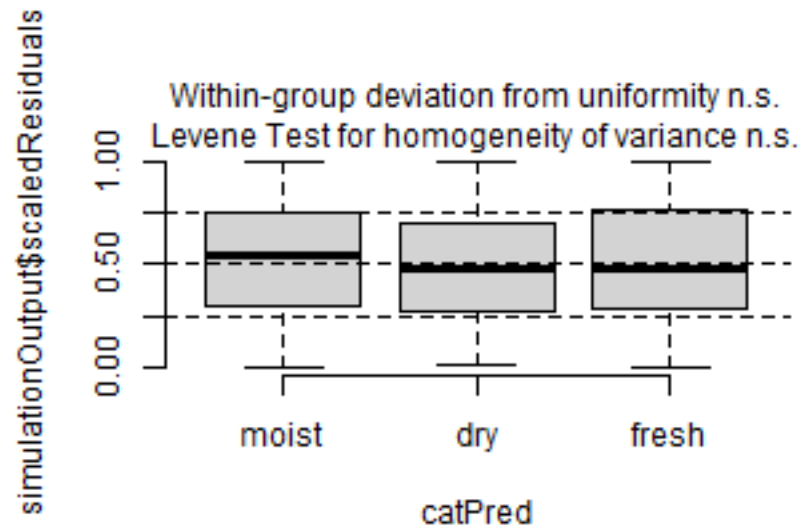
```
plotResiduals(simulation_output_2$scaledResiduals, sites$obs.year)
```



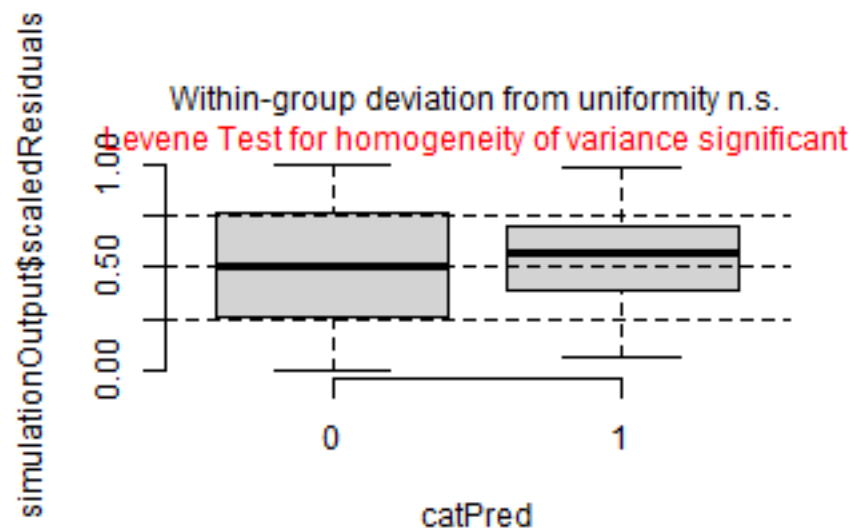
```
plotResiduals(simulation_output_1$scaledResiduals, sites$hydrology)
```



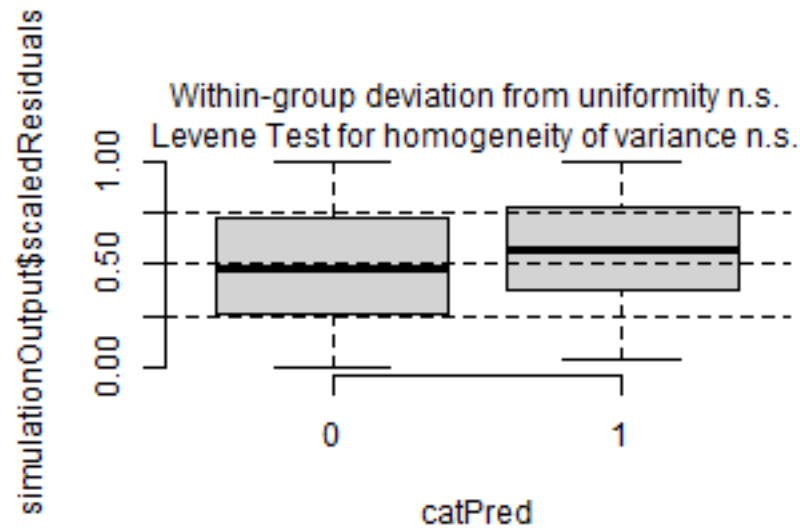
```
plotResiduals(simulation_output_2$scaledResiduals, sites$hydrology)
```



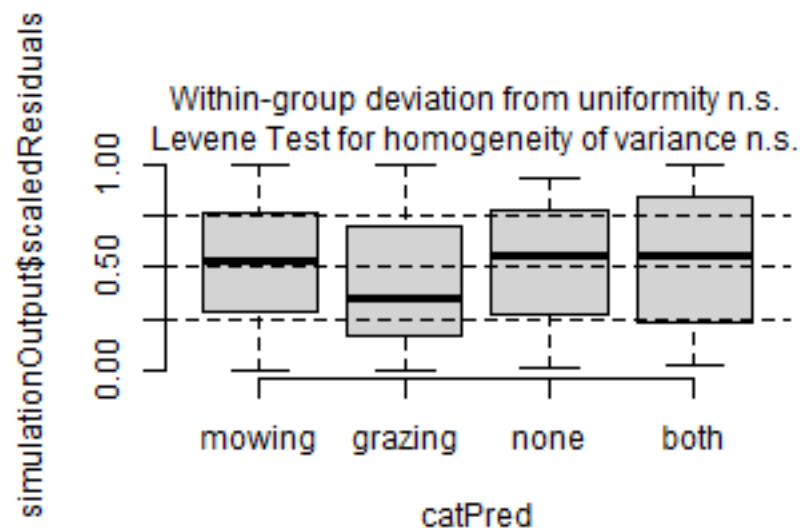
```
plotResiduals(simulation_output_1$scaledResiduals, sites$fertilized)
```



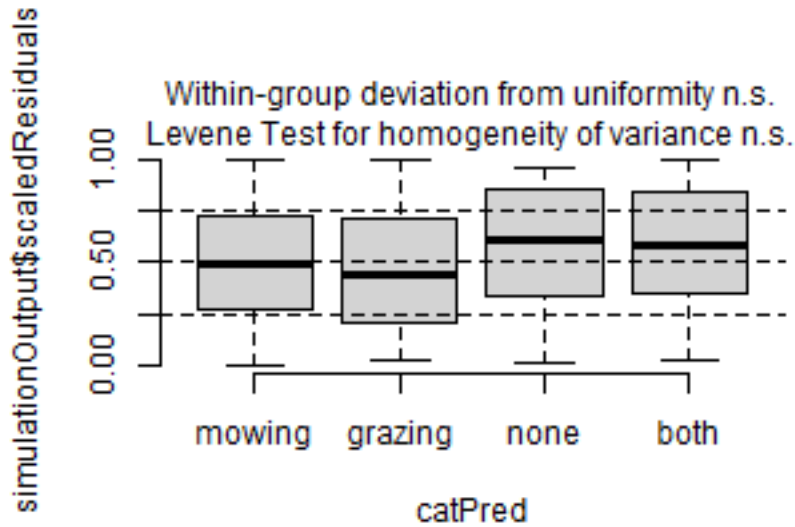
```
plotResiduals(simulation_output_2$scaledResiduals, sites$fertilized)
```



```
plotResiduals(simulation_output_1$scaledResiduals, sites$mngm.type)
```



```
plotResiduals(simulation_output_2$scaledResiduals, sites$mngm.type)
```



Check collinearity part 2 (Step 5)

Remove $VIF > 3$ or > 10 Zuur et al. 2010 Methods Ecol Evol DOI: 10.1111/j.2041-210X.2009.00001.x

```
car::vif(m_1)
```

```
##                GVIF Df  GVIF^(1/(2*Df))
## esy4           11.414767 2      1.838090
## site.type      1.371237 2      1.082127
## eco.id         1.524788 2      1.111226
## obs.year       1.020636 1      1.010265
## esy4:site.type  4.925531 4      1.220553
## esy4:eco.id     8.915162 3      1.439975
```

```
car::vif(m_2)
```

```
##                GVIF Df  GVIF^(1/(2*Df))
## esy4           3.939149 2      1.408804
## site.type      1.483165 2      1.103564
## eco.id         1.080657 2      1.019581
## obs.year       1.027422 1      1.013618
## hydrology      1.268040 2      1.061166
## esy4:site.type  4.626501 4      1.211035
```

Model comparison

R2 values

```
MuMIn::r.squaredGLMM(m_1)
##          R2m          R2c
## [1,] 0.3313799 0.7767097
MuMIn::r.squaredGLMM(m_2)
##          R2m          R2c
## [1,] 0.4233399 0.790452
```


AICc

Use AICc and not AIC since ratio $n/K < 40$ Burnham & Anderson 2002 p. 66 ISBN: 978-0-387-95364-9

```
MuMin::AICc(m_1, m_2) %>%  
  arrange(AICc)  
##      df      AICc  
## m_2 16 5538.950  
## m_1 17 5568.802
```

Predicted values

Summary table

```
car::Anova(m_2, type = 3)  
  
## Analysis of Deviance Table (Type III Wald chisquare tests)  
##  
## Response: y  
##  
##           Chisq Df Pr(>Chisq)  
## (Intercept) 1903.1430 1 < 2.2e-16 ***  
## esy4         8.7355 2  0.01268 *  
## site.type    28.3086 2  7.126e-07 ***  
## eco.id       39.4132 2  2.764e-09 ***  
## obs.year     3.0430 1  0.08109 .  
## hydrology    29.9967 2  3.064e-07 ***  
## esy4:site.type 9.0403 4  0.06010 .  
## ---  
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1  
  
summary(m_2)  
  
## Linear mixed model fit by maximum likelihood ['lmerMod']  
## Formula: y ~ esy4 * site.type + eco.id + obs.year + hydrology + (1 | id.site)  
## Data: sites  
##  
##      AIC      BIC    logLik -2*log(L)  df.resid  
##    5538    5609    -2753     5506      605  
##  
## Scaled residuals:  
##      Min       1Q   Median       3Q      Max  
## -3.1782 -0.5504 -0.0016  0.5039  3.3962  
##  
## Random effects:  
## Groups   Name      Variance Std.Dev.  
## id.site  (Intercept) 418.7    20.46  
## Residual                239.0    15.46  
## Number of obs: 621, groups: id.site, 177  
##  
## Fixed effects:  
##              Estimate Std. Error t value  
## (Intercept)    211.7360     4.8535  43.625  
## esy4R22         2.9293     2.5382   1.154  
## esy4R1A        -13.3948     5.1132  -2.620  
## site.type.L     20.8033     4.3645   4.766  
## site.type.Q      6.1834     3.1590   1.957
```

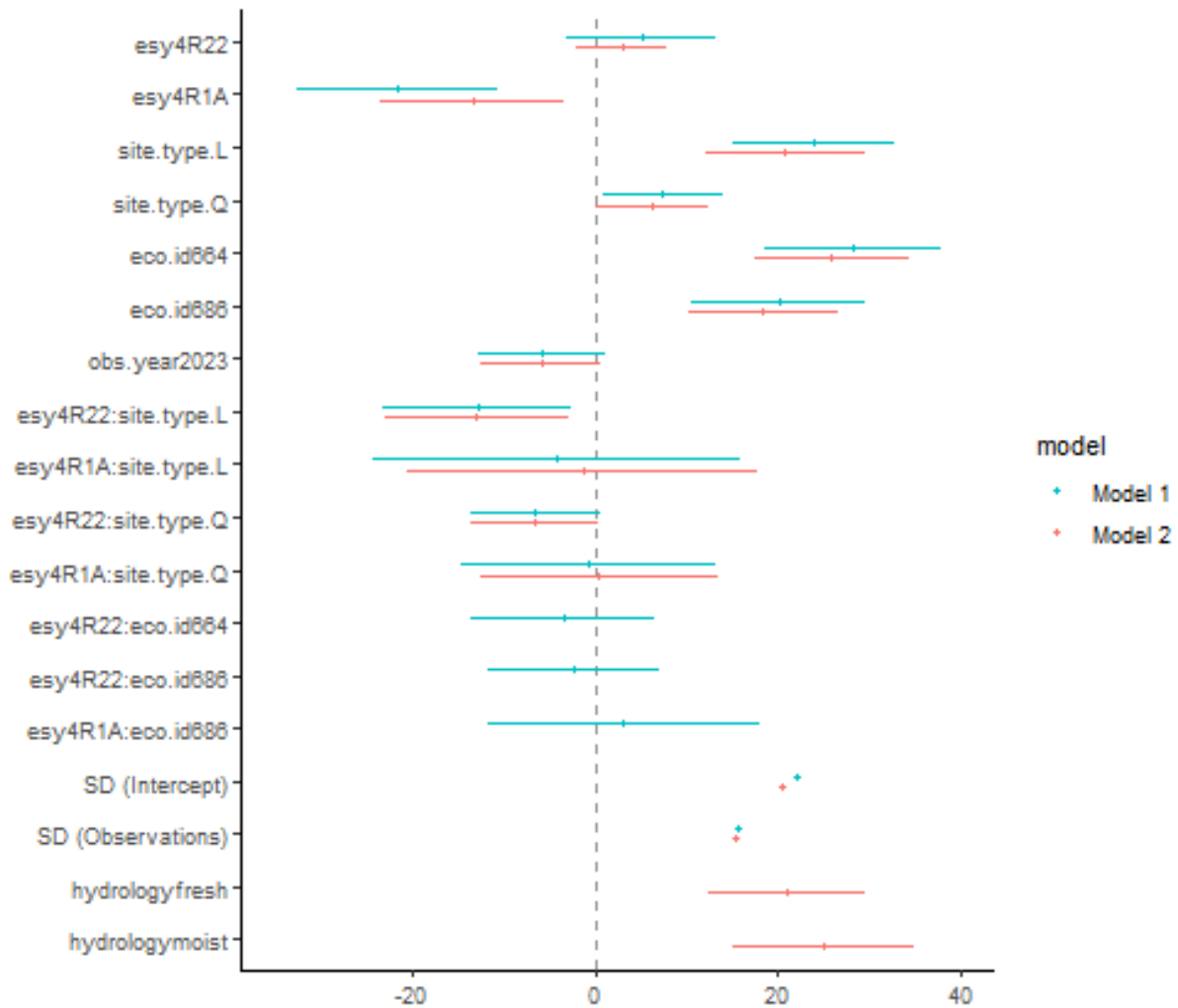
```
## eco.id664          25.8684      4.2421   6.098
## eco.id686          18.3211      4.1408   4.425
## obs.year2023       -5.9023      3.3835  -1.744
## hydrologyfresh     20.9107      4.3766   4.778
## hydrologymoist      24.9844      5.0251   4.972
## esy4R22:site.type.L -12.9476     5.1252  -2.526
## esy4R1A:site.type.L  -1.3590     9.7365  -0.140
## esy4R22:site.type.Q  -6.5424      3.4948  -1.872
## esy4R1A:site.type.Q   0.4328      6.6125   0.065

##
## Correlation matrix not shown by default, as p = 14 > 12.
## Use print(x, correlation=TRUE) or
##      vcov(x)          if you need it
```

Forest plot

```
dotwhisker::dwplot(
  list(m_1, m_2),
  ci = 0.95,
  show_intercept = FALSE,
  vline = geom_vline(xintercept = 0, colour = "grey60", linetype = 2)) +
  xlim(-35, 40) +
  theme_classic()
```

```
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
## Package 'merDeriv' needs to be installed to compute confidence intervals
##   for random effect parameters.
```



Effect sizes

Effect sizes of chosen model just to get exact values of means etc. if necessary.

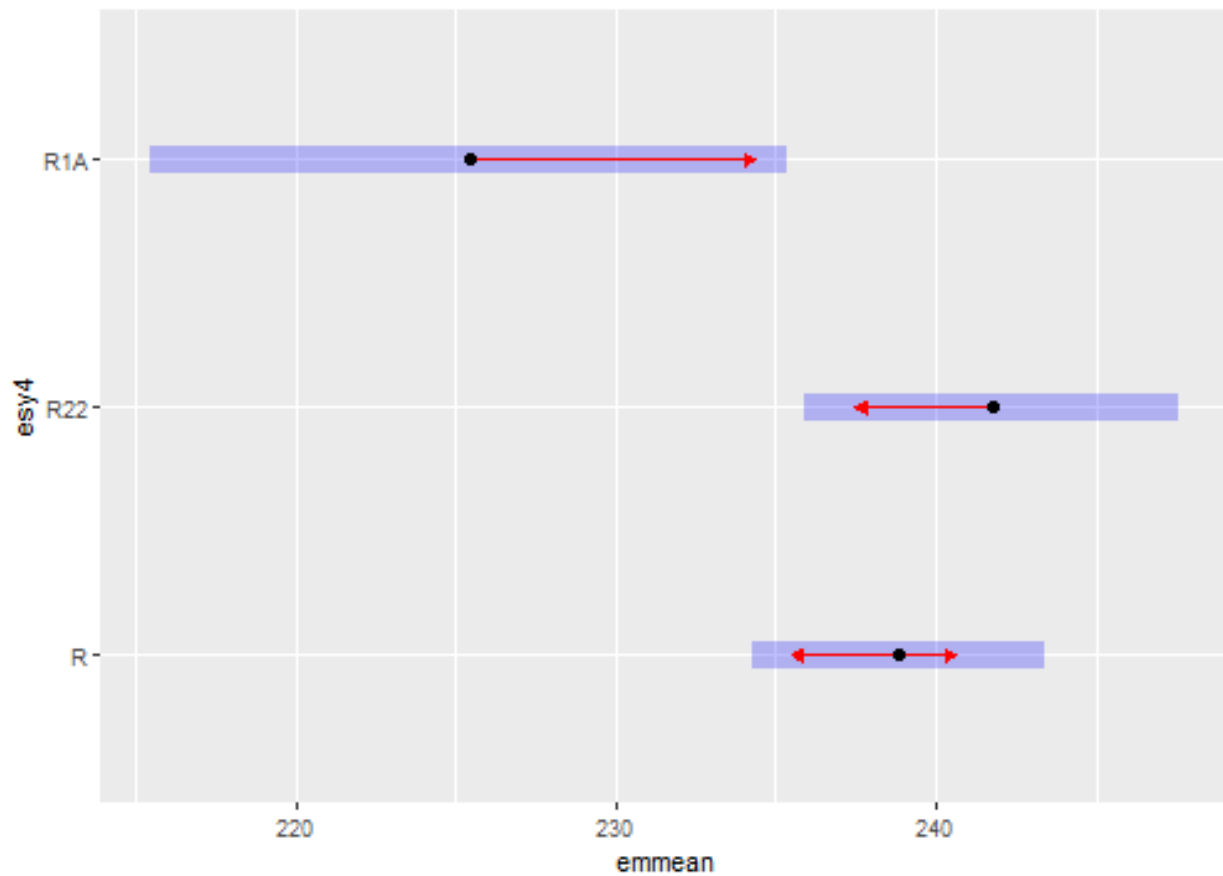
```
(emm <- emmeans(
  m_2,
  revpairwise ~ esy4,
  type = "response"
))
```

ESY EUNIS Habitat type

```
## $emmeans
## esy4 emmean SE df lower.CL upper.CL
## R      239 2.31 155      234      243
## R22    242 2.97 282      236      248
## R1A    225 5.04 316      216      235
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
```

```
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast estimate SE df t.ratio p.value
## R22 - R      2.93 2.56 556  1.143  0.4880
## R1A - R     -13.39 5.20 306 -2.574  0.0283
## R1A - R22    -16.32 5.62 344 -2.906  0.0108
##
## Results are averaged over the levels of: site.type, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
```

```
plot(emm, comparison = TRUE)
```

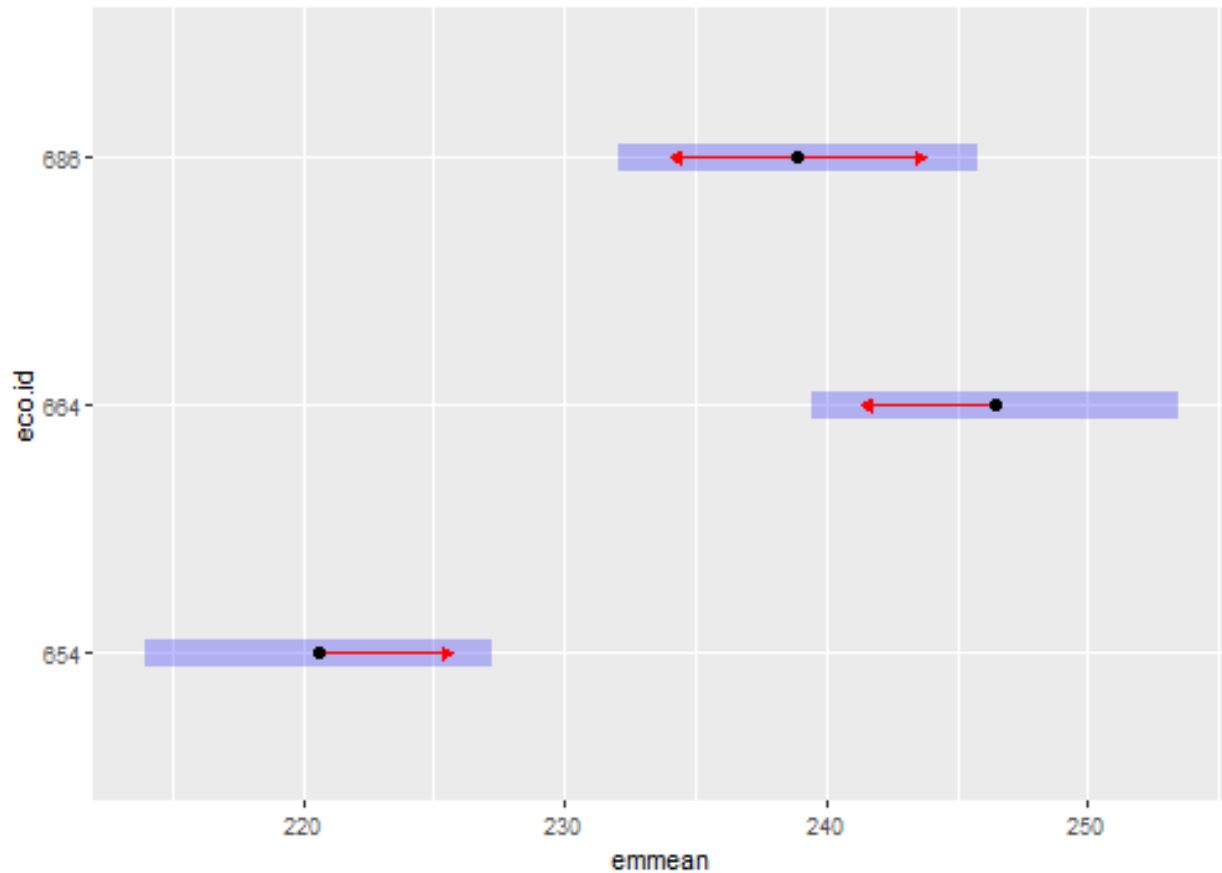


```
(emm <- emmeans(
  m_2,
  revpairwise ~ eco.id,
  type = "response"
))
```

Habitat type x Region

```
## $emmeans
## eco.id emmean SE df lower.CL upper.CL
```

```
## 654      221 3.37 153      214      227
## 664      246 3.56 226      239      253
## 686      239 3.50 186      232      246
##
## Results are averaged over the levels of: esy4, site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
## contrast      estimate    SE  df t.ratio p.value
## eco.id664 - eco.id654    25.87 4.36 186   5.933 <.0001
## eco.id686 - eco.id654    18.32 4.26 181   4.302  0.0001
## eco.id686 - eco.id664    -7.55 4.24 185  -1.779  0.1794
##
## Results are averaged over the levels of: esy4, site.type, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
plot(emm, comparison = TRUE)
```

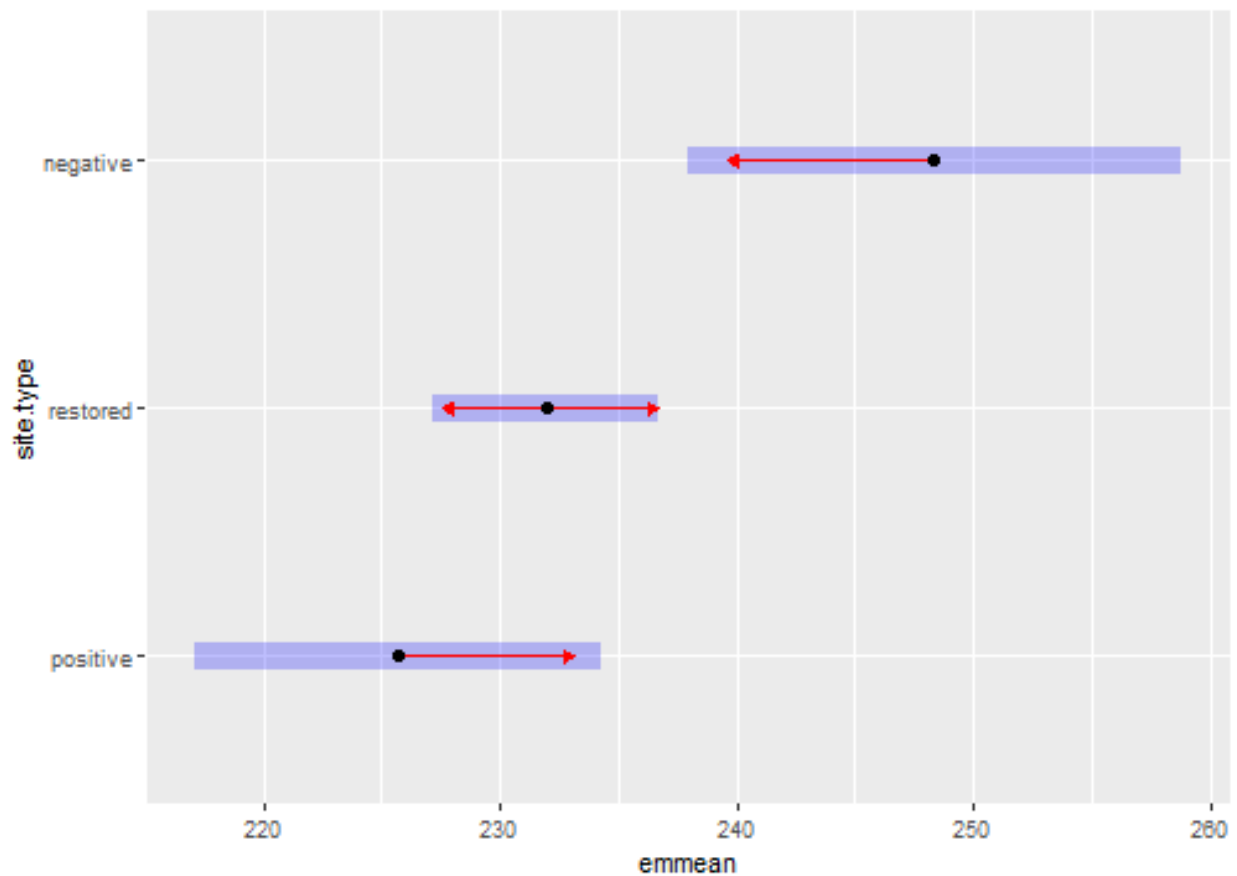


```
(emm <- emmeans(
  m_2,
  revpairwise ~ site.type,
  type = "response"
```

```
))
```

Habitat type x Site type

```
## $emmeans
##   site.type emmean    SE  df lower.CL upper.CL
##   positive    226 4.37 204     217     234
##   restored    232 2.41 128     227     237
##   negative    248 5.30 241     238     259
##
## Results are averaged over the levels of: esy4, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## Confidence level used: 0.95
##
## $contrasts
##   contrast      estimate    SE  df t.ratio p.value
##   restored - positive     6.26 4.99 184   1.254 0.4233
##   negative - positive    22.68 6.90 221   3.288 0.0034
##   negative - restored    16.42 5.74 266   2.859 0.0127
##
## Results are averaged over the levels of: esy4, eco.id, obs.year, hydrology
## Degrees-of-freedom method: kenward-roger
## P value adjustment: tukey method for comparing a family of 3 estimates
plot(emm, comparison = TRUE)
```



Session info

```
## R version 4.5.1 (2025-06-13 ucrt)
## Platform: x86_64-w64-mingw32/x64
## Running under: Windows 11 x64 (build 26100)
##
## Matrix products: default
##   LAPACK version 3.12.1
##
## locale:
## [1] LC_COLLATE=German_Germany.utf8  LC_CTYPE=German_Germany.utf8
## [3] LC_MONETARY=German_Germany.utf8 LC_NUMERIC=C
## [5] LC_TIME=German_Germany.utf8
##
## time zone: Europe/Berlin
## tzcode source: internal
##
## attached base packages:
## [1] stats      graphics  grDevices  utils      datasets  methods    base
##
## other attached packages:
## [1] emmeans_1.11.1  DHARMA_0.4.7    patchwork_1.3.0 ggbeeswarm_0.7.2
## [5] lubridate_1.9.4 forcats_1.0.0   stringr_1.5.1   dplyr_1.1.4
## [9] purrr_1.0.4     readr_2.1.5     tidyr_1.3.1     tibble_3.2.1
## [13] ggplot2_3.5.2   tidyverse_2.0.0 here_1.0.1
##
## loaded via a namespace (and not attached):
## [1] Rdpack_2.6.4      gridExtra_2.3      sandwich_3.1-1
## [4] rlang_1.1.6       magrittr_2.0.3     multcomp_1.4-28
## [7] compiler_4.5.1    mgcv_1.9-3         vctr_0.6.5
## [10] pkgconfig_2.0.3   crayon_1.5.3       fastmap_1.2.0
## [13] backports_1.5.0   labeling_0.4.3     utf8_1.2.5
## [16] ggstance_0.3.7    promises_1.3.2     rmarkdown_2.29
## [19] tzdb_0.5.0        nloptr_2.2.1       bit_4.6.0
## [22] xfun_0.52         later_1.4.2        broom_1.0.8
## [25] parallel_4.5.1    R6_2.6.1           gap.datasets_0.0.6
## [28] stringi_1.8.7     qgam_2.0.0          RColorBrewer_1.1-3
## [31] car_3.1-3         boot_1.3-31        estimability_1.5.1
## [34] Rcpp_1.0.14       iterators_1.0.14    knitr_1.50
## [37] zoo_1.8-14        parameters_0.25.0   httpuv_1.6.16
## [40] Matrix_1.7-3      splines_4.5.1       timechange_0.3.0
## [43] tidyselect_1.2.1  rstudioapi_0.17.1  abind_1.4-8
## [46] yaml_2.3.10       MuMin_1.48.11       doParallel_1.0.17
## [49] codetools_0.2-20  lattice_0.22-7      plyr_1.8.9
## [52] bayestestR_0.15.3 shiny_1.10.0        withr_3.0.2
## [55] evaluate_1.0.3    marginaleffects_0.25.1 survival_3.8-3
## [58] pillar_1.10.2     gap_1.6             carData_3.0-5
## [61] foreach_1.5.2     stats4_4.5.1        reformulas_0.4.1
## [64] insight_1.2.0     generics_0.1.4      vroom_1.6.5
## [67] rprojroot_2.0.4   hms_1.1.3           scales_1.4.0
## [70] minqa_1.2.8       xtable_1.8-4        glue_1.8.0
## [73] tools_4.5.1       data.table_1.17.2   lme4_1.1-37
## [76] mvtnorm_1.3-3     grid_4.5.1          rbibutils_2.3
## [79] datawizard_1.1.0  nlme_3.1-168        Rmisc_1.5.1
```

## [82]	performance_0.13.0	beeswarm_0.4.0	vipor_0.4.7
## [85]	Formula_1.2-5	cli_3.6.5	gtable_0.3.6
## [88]	digest_0.6.37	pbkrtest_0.5.4	TH.data_1.1-3
## [91]	farver_2.1.2	htmltools_0.5.8.1	lifecycle_1.0.4
## [94]	mime_0.13	bit64_4.6.0-1	dotwhisker_0.8.4
## [97]	MASS_7.3-65		